

Monopole, dipole, and quadrupole

□ We notice the following breakdown:

$$\begin{aligned}
 \rho'(\mathbf{x}, t)c_\infty^2 = & \overbrace{\int_S \left[\frac{\partial(\rho v_j)}{\partial \tau} \right]_{\tau=\tau^*} \frac{n_j dS(\mathbf{y})}{4\pi|\mathbf{x}-\mathbf{y}|}}^{(\rho'c_\infty^2)_{\text{monopole}}} \overbrace{- \frac{\partial}{\partial x_i} \int_S [p_{ij} + \rho v_i v_j]_{\tau=\tau^*} \frac{n_j dS(\mathbf{y})}{4\pi|\mathbf{x}-\mathbf{y}|}}^{(\rho'c_\infty^2)_{\text{dipole}}} \\
 & + \underbrace{\frac{\partial^2}{\partial x_i \partial x_j} \int_V T_{ij}(\mathbf{y}, \tau^*) \frac{dV(\mathbf{y})}{4\pi|\mathbf{x}-\mathbf{y}|}}_{(\rho'c_\infty^2)_{\text{quadrupole}}}
 \end{aligned}$$

Monopole

- ❑ Surface on which the fluid moves with a normal velocity $u_n = v_i n_i$
- ❑ Due to **unsteady mass flux** across surface
 - Notice that the mass flux is *zero* for **rigid or impermeable stationary surfaces**

$$(\rho'(\mathbf{x}, t) c_\infty^2)_{\text{monopole}} = \int_S \left[\frac{\partial(\rho u_n)}{\partial \tau} \right]_{\tau=\tau^*} \frac{dS(\mathbf{y})}{4\pi|\mathbf{x} - \mathbf{y}|}$$

- ❑ If **acoustically compact** (surface is sufficiently small, retarded time effects can be ignored), we obtain in the acoustic far-field $|\mathbf{x}| \gg |\mathbf{y}|$ that:



$$(\rho'(\mathbf{x}, t) c_\infty^2)_{\text{monopole}} = \frac{1}{4\pi|\mathbf{x}|} \left[\frac{\partial \dot{M}}{\partial \tau} \right]_{\tau=\tau^*}$$

where $\dot{M} = \int_S \rho u_n dS(\mathbf{y})$ Mass flow rate out of surface

- **Omnidirectional** source
- Driven by **net rate of mass flux through the surface** – **independent of shape!**
 - Thus, if acoustically compact, similar sound radiation is obtained from a vibrating surface of arbitrary shape or a spherical surface, given they have the same net rate of mass flux injection

Dipole

□ Let us consider the dipole term, which has terms related to the surface loading

$$(\rho'(\mathbf{x}, t)c_\infty^2)_{\text{dipole}} = -\frac{\partial}{\partial x_i} \int_S [p_{ij} + \rho v_i v_j]_{\tau=\tau^*} \frac{n_j dS(\mathbf{y})}{4\pi|\mathbf{x}-\mathbf{y}|} \Rightarrow (\rho'(\mathbf{x}, t)c_\infty^2)_{\text{dipole}} = -\int_S \frac{\partial}{\partial x_i} \left(\frac{[p_{ij} + \rho v_i v_j]_{\tau=\tau^*}}{|\mathbf{x}-\mathbf{y}|} \right) \frac{n_j dS(\mathbf{y})}{4\pi}$$

$$\Rightarrow (\rho'(\mathbf{x}, t)c_\infty^2)_{\text{dipole}} = -\int_S \left(\frac{1}{|\mathbf{x}-\mathbf{y}|} \frac{\partial}{\partial x_i} [p_{ij} + \rho v_i v_j]_{\tau=\tau^*} + [p_{ij} + \rho v_i v_j]_{\tau=\tau^*} \frac{\partial}{\partial x_i} \left[\frac{1}{|\mathbf{x}-\mathbf{y}|} \right] \right) \frac{n_j dS(\mathbf{y})}{4\pi}$$

➤ We switch the space derivative of the left term into a time derivative by using the chain rule:

$$\frac{\partial f(\tau^*)}{\partial x_i} = \frac{\partial \tau^*}{\partial x_i} \left[\frac{\partial f(\tau)}{\partial \tau} \right]_{\tau=\tau^*} \Rightarrow \frac{\partial f(\tau^*)}{\partial x_i} = \frac{-(x_i - y_i)}{|\mathbf{x}-\mathbf{y}|c_\infty} \left[\frac{\partial f(\tau)}{\partial \tau} \right]_{\tau=\tau^*}$$

$$\begin{aligned} \tau^* &= t - |\mathbf{x}-\mathbf{y}|/c_\infty \\ r &= |\mathbf{x}-\mathbf{y}| \\ \frac{\partial r}{\partial x_i} &= \frac{x_i - y_i}{|\mathbf{x}-\mathbf{y}|} \\ \frac{\partial(1/r)}{\partial x_i} &= \frac{-(x_i - y_i)}{|\mathbf{x}-\mathbf{y}|^3} \end{aligned}$$

➤ Thus, we obtain an alternate form of the dipole term:

$$(\rho'(\mathbf{x}, t)c_\infty^2)_{\text{dipole}} = \int_S \left[\underbrace{\frac{\partial(p_{ij} + \rho v_i v_j)}{\partial \tau}}_{\text{Far-field dominant}} + \underbrace{\frac{(p_{ij} + \rho v_i v_j)c_\infty}{|\mathbf{x}-\mathbf{y}|}}_{\text{Near-field dominant}} \right]_{\tau=\tau^*} \frac{(x_i - y_i)n_j dS(\mathbf{y})}{4\pi|\mathbf{x}-\mathbf{y}|^2 c_\infty}$$

Far-field
dominant

Near-field dominant

Dipole

- ❑ For **rigid or impermeable (non-porous) stationary surfaces**, we have: $\rho v_i(v_j n_j) = \rho v_i \mathbf{v} \cdot \mathbf{n} = 0$
- ❑ In most applications, the far-field sound is the main interest
- ❑ For a **rigid stationary surface, the far-field approximation** ($|\mathbf{x}| \gg |\mathbf{y}|$) yields:

$$(\rho'(\mathbf{x}, t) c_\infty^2)_{\text{dipole}} \approx \frac{x_i}{4\pi |\mathbf{x}|^2 c_\infty} \int_S \left[\frac{\partial(p_{ij} n_j)}{\partial \tau} \right]_{\tau=\tau^*} dS(\mathbf{y})$$

For a rigid surface, **ONLY surface loading causes sound**

- Typically, the viscous term σ_{ij} is *neglected*. If not, one need to evaluate the volume sources in the boundary layer close to the wall
- ❑ For **acoustically compact surface** (neglect retarded time effects over the surface):

$$(\rho'(\mathbf{x}, t) c_\infty^2)_{\text{dipole}} \approx \frac{x_i}{4\pi |\mathbf{x}|^2 c_\infty} \left[\frac{\partial F_i}{\partial \tau} \right]_{\tau=\tau^*}$$

where $F_i(\tau) = \int_S p_{ij} n_j dS(\mathbf{y})$ Net force applied to fluid by surface (incl. viscous drag)

- Notice the **cosine directivity** ($x_i/|\mathbf{x}|$) → *similar* to a **compact dipole source**

Quadrupole

$$\begin{aligned}\tau^* &= t - |\mathbf{x} - \mathbf{y}|/c_\infty \\ r &= |\mathbf{x} - \mathbf{y}| \\ \frac{\partial r}{\partial x_i} &= \frac{x_i - y_i}{|\mathbf{x} - \mathbf{y}|} \\ \frac{\partial(1/r)}{\partial x_i} &= \frac{-(x_i - y_i)}{|\mathbf{x} - \mathbf{y}|^3}\end{aligned}$$

□ Let us consider the quadrupole term, which is related to the volume integral:

$$(\rho'(\mathbf{x}, t) c_\infty^2)_{\text{quad}} = \frac{\partial^2}{\partial x_i \partial x_j} \int_V T_{ij}(\mathbf{y}, \tau^*) \frac{dV(\mathbf{y})}{4\pi |\mathbf{x} - \mathbf{y}|} = \int_V \frac{\partial}{\partial x_i} \left\{ \frac{\partial}{\partial x_j} \left(\frac{[T_{ij}]_{\tau=\tau^*}}{r} \right) \right\} \frac{dV(\mathbf{y})}{4\pi} \Rightarrow (\rho'(\mathbf{x}, t) c_\infty^2)_{\text{quad}} = \int_V \frac{\partial}{\partial x_i} \left\{ \frac{1}{r} \left[\frac{\partial T_{ij}}{\partial x_j} \right]_{\tau=\tau^*} + [T_{ij}]_{\tau=\tau^*} \frac{\partial}{\partial x_j} \left(\frac{1}{r} \right) \right\} \frac{dV(\mathbf{y})}{4\pi}$$

➤ As before, we switch the space derivatives to derivatives over source time:

$$\frac{\partial f(\tau^*)}{\partial x_i} = \frac{\partial \tau^*}{\partial x_i} \left[\frac{\partial f(\tau)}{\partial \tau} \right]_{\tau=\tau^*} \quad \text{where} \quad \frac{\partial \tau^*}{\partial x_i} = \frac{-(x_i - y_i)}{r c_\infty}$$

$$\Rightarrow (\rho'(\mathbf{x}, t) c_\infty^2)_{\text{quad}} = \int_V \frac{\partial}{\partial x_i} \left\{ \frac{1}{r} \left[\frac{\partial \tau^*}{\partial x_j} \frac{\partial T_{ij}}{\partial \tau} \right]_{\tau=\tau^*} + [T_{ij}]_{\tau=\tau^*} \frac{\partial}{\partial x_j} \left(\frac{1}{r} \right) \right\} \frac{dV(\mathbf{y})}{4\pi} \Rightarrow (\rho'(\mathbf{x}, t) c_\infty^2)_{\text{quad}} = \int_V \left\{ \left[\frac{\partial T_{ij}}{\partial \tau} \right]_{\tau=\tau^*} \frac{\partial}{\partial x_i} \left(\frac{1}{r} \frac{\partial \tau^*}{\partial x_j} \right) + \frac{1}{r} \frac{\partial \tau^*}{\partial x_j} \frac{\partial \tau^*}{\partial x_i} \left[\frac{\partial^2 T_{ij}}{\partial \tau^2} \right]_{\tau=\tau^*} \right. \\ \left. + \frac{\partial \tau^*}{\partial x_i} \left[\frac{\partial T_{ij}}{\partial \tau} \right]_{\tau=\tau^*} \frac{\partial}{\partial x_j} \left(\frac{1}{r} \right) + [T_{ij}]_{\tau=\tau^*} \frac{\partial^2}{\partial x_i \partial x_j} \left(\frac{1}{r} \right) \right\} \frac{dV(\mathbf{y})}{4\pi}$$

➤ Thus, we obtain a quadrupole term:

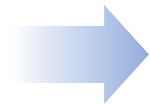
$$\Rightarrow (\rho'(\mathbf{x}, t) c_\infty^2)_{\text{quad.}} = \int_V \left\{ \frac{1}{r} \frac{\partial \tau^*}{\partial x_i} \frac{\partial \tau^*}{\partial x_j} \left[\frac{\partial^2 T_{ij}}{\partial \tau^2} \right]_{\tau=\tau^*} + \left[\frac{\partial T_{ij}}{\partial \tau} \right]_{\tau=\tau^*} \left(\frac{\partial \tau^*}{\partial x_i} \frac{\partial}{\partial x_j} \left(\frac{1}{r} \right) + \frac{\partial}{\partial x_i} \left(\frac{1}{r} \frac{\partial \tau^*}{\partial x_j} \right) \right) + [T_{ij}]_{\tau=\tau^*} \frac{\partial^2}{\partial x_i \partial x_j} \left(\frac{1}{r} \right) \right\} \frac{dV(\mathbf{y})}{4\pi}$$

Quadrupole

$$\begin{aligned}\tau^* &= t - |\mathbf{x} - \mathbf{y}|/c_\infty \\ r &= |\mathbf{x} - \mathbf{y}| \\ \frac{\partial \tau^*}{\partial x_i} &= \frac{-(x_i - y_i)}{rc_\infty}\end{aligned}$$

- In the **acoustic far-field** ($|\mathbf{x}| \gg |\mathbf{y}|$), where Lighthill's analogy applies, we can simplify the term by **dropping all terms of order $1/r^2$ and $1/r^3$** :

$$(\rho'(\mathbf{x}, t)c_\infty^2)_{\text{quadrupole}} \approx \int_V \left\{ \frac{1}{r} \frac{\partial \tau^*}{\partial x_i} \frac{\partial \tau^*}{\partial x_j} \left[\frac{\partial^2 T_{ij}}{\partial \tau^2} \right]_{\tau=\tau^*} \right\} \frac{dV(\mathbf{y})}{4\pi} \quad \text{where} \quad r \approx |\mathbf{x}| \quad \frac{\partial \tau^*}{\partial x_i} \approx \frac{-x_i}{|\mathbf{x}|c_\infty} \quad \frac{\partial \tau^*}{\partial x_j} \approx \frac{-x_j}{|\mathbf{x}|c_\infty}$$



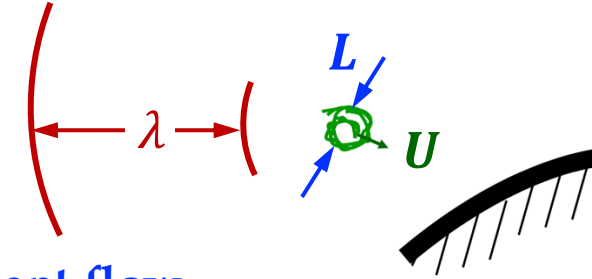
$$(\rho'(\mathbf{x}, t)c_\infty^2)_{\text{quadrupole}} \approx \frac{x_i x_j}{4\pi |\mathbf{x}|^3 c_\infty^2} \int_V \left[\frac{\partial^2 T_{ij}}{\partial \tau^2} \right]_{\tau=\tau^*} dV(\mathbf{y})$$

- If volume of turbulent flow region is small compared to the acoustic wavelength, that is **acoustically compact**, we can ignore effects of retarded time across the volume, leading to:

$$(\rho'(\mathbf{x}, t)c_\infty^2)_{\text{quadrupole}} \approx \frac{x_i x_j}{4\pi |\mathbf{x}|^3 c_\infty^2} \left[\int_V \frac{\partial^2 T_{ij}}{\partial \tau^2} dV(\mathbf{y}) \right]_{\tau=\tau^*}$$

- Notice the **directivity** ($x_i x_j / |\mathbf{x}|^2$) → *similar* to a **compact quadrupole source**

Scaling laws



❑ As shown by Lighthill's analogy, **any aeroacoustic source is driven by a turbulent flow**

❑ Consider a turbulent flow (source) region with the following scales:

- Length scale (e.g., vortex diameter) - L
- Mean flow speed - U
- Time scale - $\tau \propto L/U$

❑ Based on these scales, we can estimate the following quantities:

$$\left\{ \begin{array}{l} \frac{\partial}{\partial x_i} \propto \frac{1}{L} \quad , \quad \int dV \propto L^3 \\ F_i \propto \rho_\infty U^2 S \quad , \quad \frac{\partial F_i}{\partial \tau} \propto \frac{\rho_\infty U^3 S}{L} \\ S \text{ is the surface area of body} \\ T_{ij} \propto \rho_\infty U^2 \end{array} \right.$$

❑ Recall, **acoustic intensity in the far-field**: $I_r = \frac{p_{\text{rms}}^2}{\rho_\infty c_\infty}$

❑ Thus, in the **far-field of acoustically compact flow sources**:

$$(I_r)_{\text{monopole}} \propto \frac{\rho_\infty U^4 S^2}{(4\pi|\mathbf{x}|)^2 L^2 c_\infty}$$

$$(I_r)_{\text{dipole}} \propto \frac{\rho_\infty U^6 S^2 \cos^2 \theta}{(4\pi|\mathbf{x}|)^2 c_\infty^3 L^2}$$

θ is the observer angle relative to the dipole axis

$$(I_r)_{\text{quad.}} \propto \frac{\rho_\infty U^8 L^2}{(4\pi|\mathbf{x}|)^2 c_\infty^5} \left(\frac{x_i x_j}{|\mathbf{x}|^2} \right)^2$$

Scaling laws

Compact sources

$$\text{Mach} \quad M = \frac{U}{c_\infty}$$

□ Monopole (4th power law)

$$(I_r)_{\text{monopole}} \propto \frac{\rho_\infty U^4 S^2}{(4\pi|\mathbf{x}|)^2 L^2 c_\infty}$$

□ Dipole (6th power law)

$$(I_r)_{\text{dipole}} \propto \frac{\rho_\infty U^6 S^2 \cos^2 \theta}{(4\pi|\mathbf{x}|)^2 c_\infty^3 L^2} = M^3 U^3$$

□ Quadrupole (8th power law)

$$(I_r)_{\text{quad.}} \propto \frac{\rho_\infty U^8 L^2}{(4\pi|\mathbf{x}|)^2 c_\infty^5} \left(\frac{x_i x_j}{|\mathbf{x}|^2} \right)^2 = M^5 U^3$$

$$\frac{(I_r)_{\text{quad.}}}{(I_r)_{\text{dipole}}} \sim M^2$$

for $M \ll 1$

$$(I_r)_{\text{dipole}} \gg (I_r)_{\text{quad.}}$$

**Non-acoustically compact source has
reduced noise radiation efficiency**
(due to interference between different parts of
the source region)

Scaling laws

Most powerful result from Lighthill's analogy

$$(I_r)_{\text{quad.}} \propto \frac{\rho_{\infty} U^8 L^2}{(4\pi |\mathbf{x}|)^2 c_{\infty}^5} \left(\frac{x_i x_j}{|\mathbf{x}|^2} \right)^2 = M^5 U^3$$

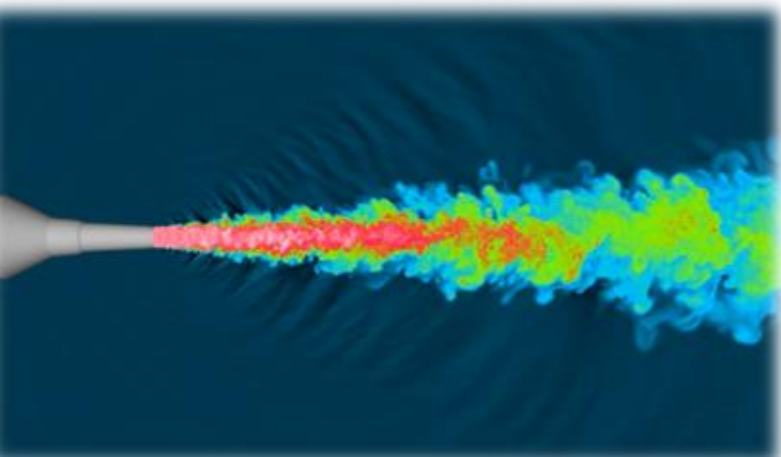
Noise from *turbulence* in the flow scales with U^8 !

□ Example

- The exit velocity from a jet is reduced by 30%. What is the change in acoustic intensity in the far-field?

$$\Delta \text{SPL} = 10 \log_{10} \left(\frac{U_{\text{new}}^8}{U_{\text{old}}^8} \right) \Rightarrow \Delta \text{SPL} = 10 \log_{10} \left(\frac{(0.7U)^8}{U^8} \right)$$

$$\Rightarrow \Delta \text{SPL} = 80 \log_{10}(0.7) = -12.4 \text{ dB} !!!$$



Scaling laws

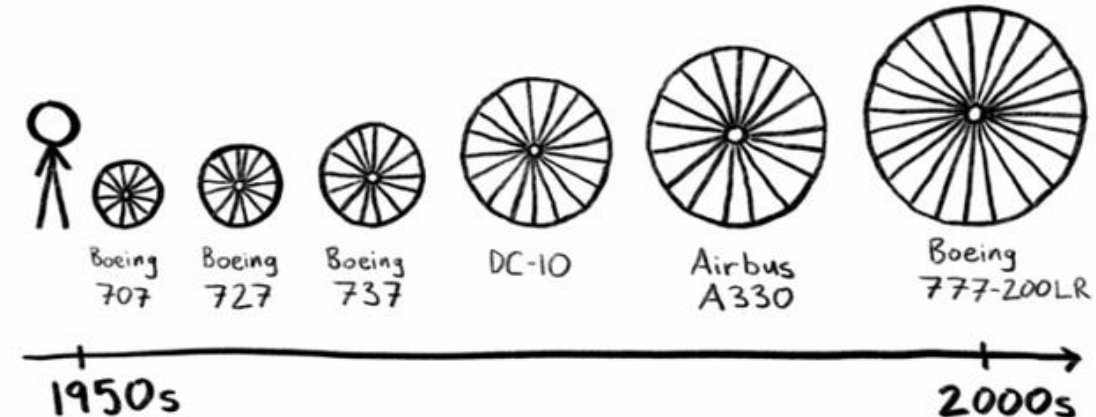
Most powerful result from Lighthill's analogy

Noise from *turbulence in the flow* scales with U^8 !

- ❑ A decrease in jet exit velocity is also accompanied by a reduction in thrust, which often limits the noise reduction that can be achieved
- ❑ The thrust is proportional to $T \propto \rho_{\infty} U^2 A$, where A is the exit area of the jet

For a given thrust, increasing the jet diameter (or exit area) can *reduce* the exit velocity, leading to **noise reduction**

- ❑ This is one of the reasons why many modern commercial aircraft have *large-diameter engines*...



Scaling laws

Most powerful result from Lighthill's analogy

Noise from *turbulence* in the flow scales with U^8 !



Scaling laws

Most powerful result from Lighthill's analogy

Noise from turbulence in the flow scales with U^8 !

□ Example 2

- For the same total thrust, would you prefer a twin-engine aircraft or a four-engine aircraft?
- We know that the thrust from ONE engine is proportional to $T \propto \rho_\infty U^2 D^2$, where D is the engine diameter
 - Since the TOTAL thrust T_{tot} is the same for both aircraft, we can write:

$$T_{\text{tot}} = 2\rho_\infty U_2^2 D_2^2 = 4\rho_\infty U_4^2 D_4^2 \quad \Rightarrow \quad \frac{U_2}{U_4} = \sqrt{2} \frac{D_4}{D_2}$$

- And the total acoustic intensity the jets is:

$$I_2 = 2U_2^8 D_2^2 \quad I_4 = 4U_4^8 D_4^2$$

- Thus, the ratio of the total noise is:

$$\frac{I_2}{I_4} = \frac{1}{2} \left(\frac{U_2}{U_4} \right)^8 \left(\frac{D_2}{D_4} \right)^2 \Rightarrow \frac{I_2}{I_4} = \frac{1}{2} \left(\sqrt{2} \frac{D_4}{D_2} \right)^8 \left(\frac{D_2}{D_4} \right)^2 \Rightarrow \boxed{\frac{I_2}{I_4} = 8 \left(\frac{D_4}{D_2} \right)^6} \xrightarrow{D_2=2D_4} \frac{I_2}{I_4} = \frac{1}{8} \Rightarrow \Delta\text{SPL} = 10\log_{10} \left(\frac{I_2}{I_4} \right) = -9 \text{ dB}$$

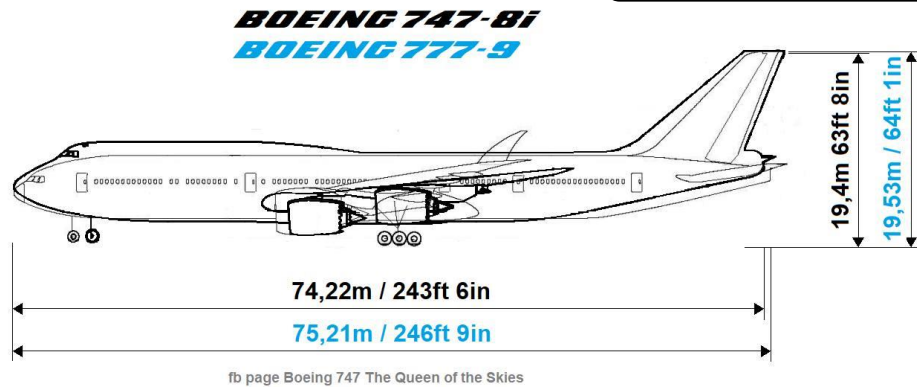
$$(I_r)_{\text{quad.}} \propto \frac{\rho_\infty U^8 L^2}{(4\pi|\mathbf{x}|)^2 c_\infty^5} \left(\frac{x_i x_j}{|\mathbf{x}|^2} \right)^2$$

- To prefer a twin-engine aircraft, we need that $\frac{I_2}{I_4} = 8 \left(\frac{D_4}{D_2} \right)^6 < 1 \Rightarrow \boxed{D_2 > \sqrt{2} D_4}$

Scaling laws

Most powerful result from Lighthill's analogy

Noise from *turbulence in the flow* scales with U^8 !



$$T_{777,\text{tot}} = 210,000 \text{ lb}_f$$

$$D_{777} = 4.17 \text{ m}$$

$$T_{747,\text{tot}} = 266,000 \text{ lb}_f$$

$$D_{747} = 2.84 \text{ m}$$

$$\frac{D_{777}}{D_{747}} = 1.47$$

$$\frac{T_{777,\text{tot}}}{T_{747,\text{tot}}} = \frac{2\rho_\infty U_{777}^2 D_{777}^2}{4\rho_\infty U_{747}^2 D_{747}^2} = \frac{210,000}{266,000} \Rightarrow \frac{U_{777}^2 D_{777}^2}{U_{747}^2 D_{747}^2} = 1.6$$

$$\Rightarrow \frac{U_{777}}{U_{747}} = 0.86$$

$$\frac{I_{777}}{I_{747}} \Rightarrow \frac{I_{777}}{I_{747}} = \frac{1}{2} (0.86)^8 (1.47)^2 = 0.31$$

$$\Rightarrow \Delta \text{SPL} = 10 \log_{10} \left(\frac{I_{777}}{I_{747}} \right) = -5 \text{ dB}$$

Scaling laws

Most powerful result from Lighthill's analogy

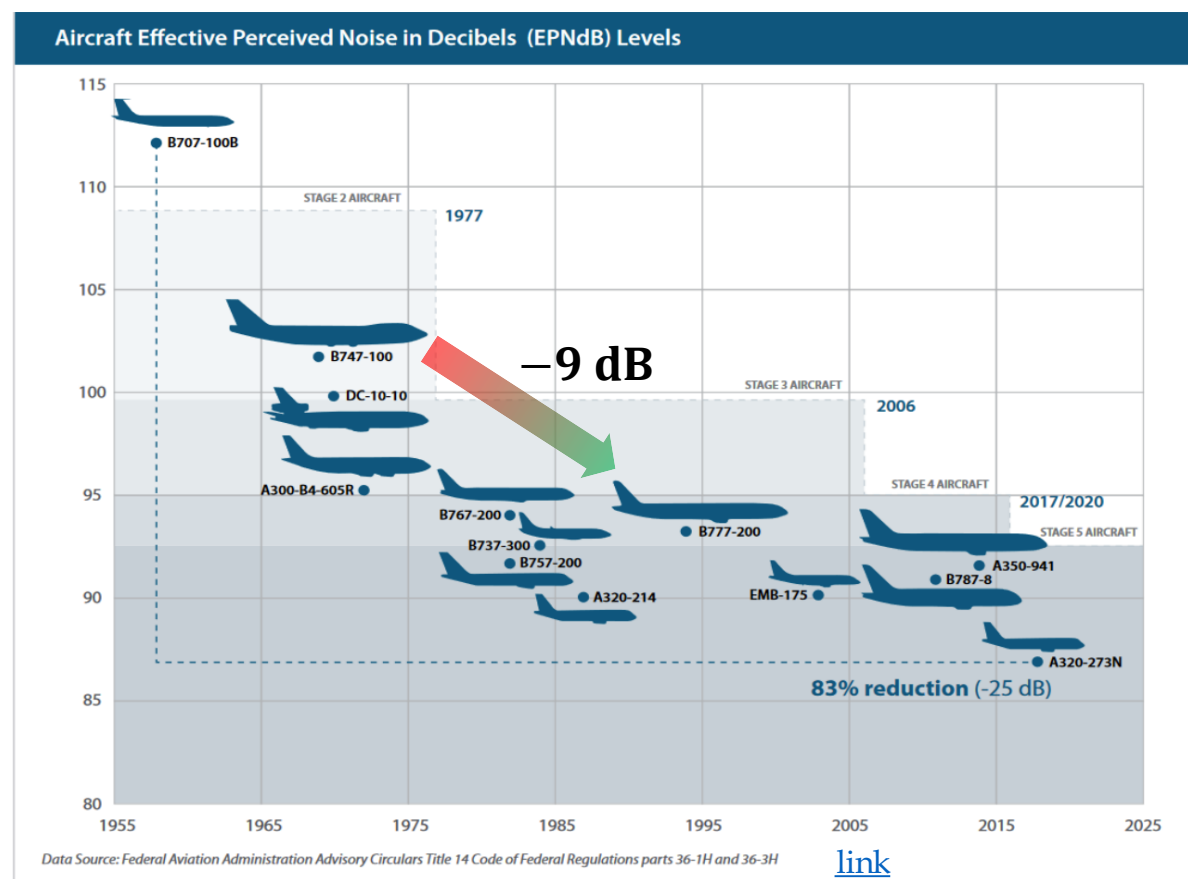
Noise from *turbulence* in the flow scales with U^8 !



Scaling laws

Most powerful result from Lighthill's analogy

Noise from *turbulence* in the flow scales with U^8 !



Wavenumber and Fourier transforms

- Wavenumber methods are very important in aeroacoustics because they directly relate the source fluctuations to the propagating waves in the acoustic far field
- Recall the Fourier transform with respect to time and the inverse transform:

$$\boxed{\tilde{f}(\omega) = \frac{1}{2\pi} \int_{-T}^{+T} f(t) e^{i\omega t} dt} \quad \boxed{f(t) = \int_{-\infty}^{+\infty} \tilde{f}(\omega) e^{-i\omega t} d\omega} \quad \text{inverse}$$

➤ $f(t)$ is a piecewise continuously differentiable function, T tends to infinity, ω is the angular frequency, and $\tilde{f}(\omega)$ is the frequency content of the acoustic field generated by a turbulent flow

- Consider the **far-field sound generated by a quadrupole** (turbulence without scattering surfaces):

$$(\rho'(\mathbf{x}, t) c_{\infty}^2)_{\text{quadrupole}} \approx \frac{x_i x_j}{4\pi |\mathbf{x}|^3 c_{\infty}^2} \int_V \left[\frac{\partial^2 T_{ij}(\mathbf{y}, \tau)}{\partial \tau^2} \right]_{\tau=\tau^*} dV(\mathbf{y}) \quad \longleftarrow \quad T_{ij}(\mathbf{y}, \tau) = \int_{-\infty}^{+\infty} \tilde{T}_{ij}(\mathbf{y}, \omega) e^{-i\omega \tau} d\omega$$

- Introducing the inverse Fourier transform to T_{ij} and evaluating τ at the correct retarded time τ^* :

$$\Rightarrow (\rho'(\mathbf{x}, t) c_{\infty}^2)_{\text{quadrupole}} \approx \int_{-\infty}^{+\infty} \left\{ \frac{-\omega^2 x_i x_j}{4\pi |\mathbf{x}|^3 c_{\infty}^2} \int_V \tilde{T}_{ij}(\mathbf{y}, \omega) e^{ikr} dV(\mathbf{y}) \right\} e^{-i\omega t} d\omega \quad \begin{array}{l} r = |\mathbf{x} - \mathbf{y}| \\ \tau^* = t - r/c_0 \end{array}$$

Wavenumber and Fourier transforms

Quadrupole

□ Then, taking the Fourier transformation of the previous result with respect to time gives:

$$(\rho'(\mathbf{x}, t) c_\infty^2)_{\text{quadrupole}} \approx \int_{-\infty}^{+\infty} \left\{ \frac{-\omega^2 x_i x_j}{4\pi |\mathbf{x}|^3 c_\infty^2} \int_V \tilde{T}_{ij}(\mathbf{y}, \omega) e^{ikr} dV(\mathbf{y}) \right\} e^{-i\omega t} d\omega \quad \longleftarrow \quad f(t) = \int_{-\infty}^{+\infty} \tilde{f}(\omega) e^{-i\omega t} d\omega$$



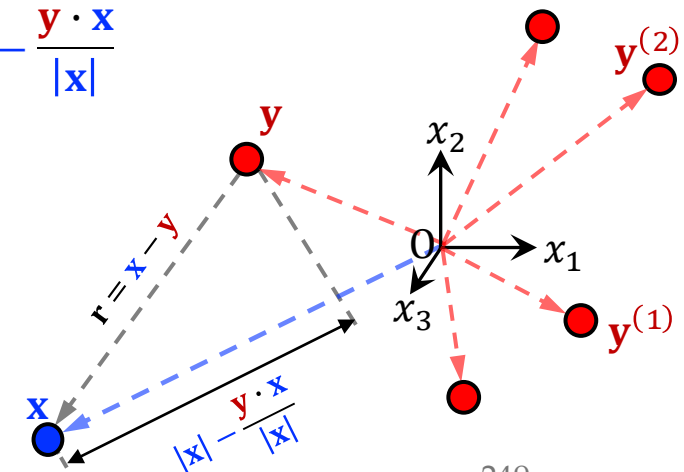
$$(\tilde{\rho}(\mathbf{x}, \omega) c_\infty^2)_{\text{quadrupole}} \approx \frac{-\omega^2 x_i x_j}{4\pi |\mathbf{x}|^3 c_\infty^2} \int_V \tilde{T}_{ij}(\mathbf{y}, \omega) e^{ikr} dV(\mathbf{y})$$

➤ Note that all the retarded time effects are now accounted for by e^{ikr} , helping us avoid the rather complicated evaluation of emission time at the source

□ In the far-field approximation ($|\mathbf{x}| \gg |\mathbf{y}|$), we can write: $r = |\mathbf{x} - \mathbf{y}| \approx |\mathbf{x}| - \frac{\mathbf{y} \cdot \mathbf{x}}{|\mathbf{x}|}$



$$(\tilde{\rho}(\mathbf{x}, \omega) c_\infty^2)_{\text{quadrupole}} \approx \frac{-\omega^2 x_i x_j e^{ik|\mathbf{x}|}}{4\pi |\mathbf{x}|^3 c_\infty^2} \int_V \tilde{T}_{ij}(\mathbf{y}, \omega) e^{-ik\mathbf{y} \cdot \mathbf{x}/|\mathbf{x}|} dV(\mathbf{y})$$



Wavenumber and Fourier transforms

Dipole

- A similar approach can be used for **dipole** surface sources...
- Taking the Fourier transform with respect to time and applying the **far-field approximation** yields:

$$(\rho'(\mathbf{x}, t)c_\infty^2)_{\text{dipole}} \approx \frac{x_i}{4\pi|\mathbf{x}|^2 c_\infty} \int_S \left[\frac{\partial(p_{ij}n_j)}{\partial \tau} \right]_{\tau=\tau^*} dS(\mathbf{y})$$

$r \approx |\mathbf{x}| - \frac{\mathbf{y} \cdot \mathbf{x}}{|\mathbf{x}|} \quad \tau^* = t - r/c_0$
 $p_{ij}(\mathbf{y}, \tau) = \int_{-\infty}^{+\infty} \tilde{p}_{ij}(\mathbf{y}, \omega) e^{-i\omega\tau} d\omega$



$$(\tilde{\rho}(\mathbf{x}, \omega)c_\infty^2)_{\text{dipole}} \approx \frac{-i\omega x_i e^{ik|\mathbf{x}|}}{4\pi|\mathbf{x}|^2 c_\infty} \int_S \tilde{p}_{ij}(\mathbf{y}, \omega) n_j e^{-ik\mathbf{y} \cdot \mathbf{x}/|\mathbf{x}|} dS(\mathbf{y})$$

- For an **acoustically compact surface** ($k|\mathbf{y}| \ll 1$), the source simplifies to the Fourier transform of the force exerted by the body on the fluid:

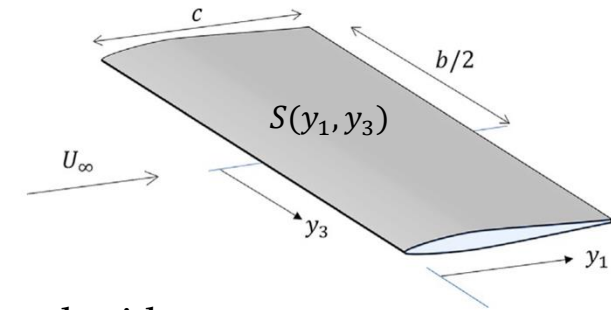
$$(\tilde{\rho}(\mathbf{x}, \omega)c_\infty^2)_{\text{dipole}} \approx \frac{-i\omega x_i e^{ik|\mathbf{x}|} \tilde{F}_i(\omega)}{4\pi|\mathbf{x}|^2 c_\infty}$$

where $\tilde{F}_i(\omega) = \int_S \tilde{p}_{ij}(\mathbf{y}, \omega) n_j dS(\mathbf{y})$

$$e^{-ik\mathbf{y} \cdot \mathbf{x}/|\mathbf{x}|} \approx 1 - ik|\mathbf{y}| \cos \theta \approx 1$$

Wavenumber and Fourier transforms

Dipole



□ Application to thin airfoils, assuming $\sigma_{ij} \approx 0$

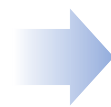
- Consider a flat plate airfoil with chord c , aligned with the y_1 -axis, and span b aligned with y_3
- Surface integral can be expressed in terms of pressure jump Δp across airfoil planform, in the $y_2 = 0$ plane:

$$[\Delta \tilde{p} n_2 dy_1 dy_3]_{\text{planform}} = [\tilde{p} n_i dS]_{\text{upper}} - [\tilde{p} n_i dS]_{\text{lower}}$$

- Applying the above to the **far-field approximation of a dipole source** results in:

$$(\tilde{\rho}(\mathbf{x}, \omega) c_\infty^2)_{\text{dipole}} \approx \frac{-i\omega x_2 e^{ik|\mathbf{x}|}}{4\pi |\mathbf{x}|^2 c_\infty} \int_0^c \int_{-b/2}^{b/2} \Delta \tilde{p}(\mathbf{y}, \omega) e^{-ikx_1 y_1/|\mathbf{x}| - ikx_3 y_3/|\mathbf{x}|} dy_1 dy_3$$

Pressure jump is zero on all parts that are *not* on the airfoil, thus integral limits can be extended to R_∞



$$(\tilde{\rho}(\mathbf{x}, \omega) c_\infty^2)_{\text{dipole}} \approx \frac{-i\pi\omega x_2 \Delta \tilde{p}(k_1^{(o)}, k_3^{(o)}, \omega) e^{ik|\mathbf{x}|}}{c_\infty |\mathbf{x}|^2}$$

$$\Delta \tilde{p}(k_1, k_3, \omega) = \frac{1}{(2\pi)^2} \int_{-R_\infty}^{R_\infty} \int_{-R_\infty}^{R_\infty} \Delta \tilde{p}(\mathbf{y}, \omega) e^{-ik_1 y_1 - ik_3 y_3} dy_1 dy_3$$

2D wavenumber frequency transform
(chordwise-spanwise)
of the pressure jump over the airfoil planform

$$k_i^{(o)} = \frac{\omega x_i}{c_\infty |\mathbf{x}|} = k \frac{x_i}{|\mathbf{x}|}$$

Acoustic wavenumber in the observer direction

$$\mathbf{k}^{(o)} = (k_1^{(o)}, k_2^{(o)}, k_3^{(o)}) = k \frac{\mathbf{x}}{|\mathbf{x}|}$$

Acoustic wavevector in the observer direction

Wavenumber and Fourier transforms

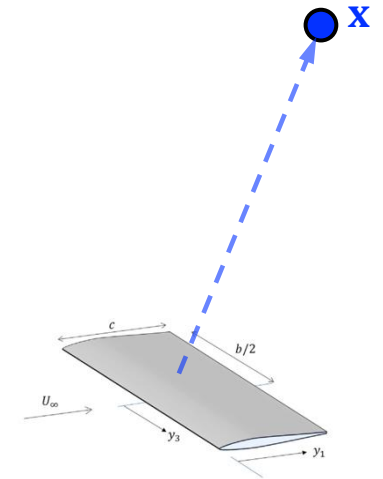
Dipole

□ Application to thin airfoils

$$k_i^{(o)} = \frac{\omega x_i}{c_\infty |\mathbf{x}|} = k \frac{x_i}{|\mathbf{x}|}$$

$$(\tilde{p}(\mathbf{x}, \omega) c_\infty^2)_{\text{dipole}} \approx \frac{-i\pi\omega x_2 \Delta\tilde{p}(k_1^{(o)}, k_3^{(o)}, \omega) e^{ik|\mathbf{x}|}}{c_\infty |\mathbf{x}|^2}$$

For a given frequency ω and observation direction $\mathbf{x}/|\mathbf{x}|$, the **far-field acoustic pressure** is proportional to the **wavenumber transform** of Δp evaluated at a *single* wavenumber pair $(k_1^{(o)}, k_3^{(o)})$ with pressure disturbances convecting at c_∞



- Other wavenumber components of the surface pressure do contribute *locally* near the airfoil, but **cancel out** in the far-field by *destructive interference*
- Disturbances that convect over the surface at *subsonic* speeds will not couple with the acoustic far-field
 - These correspond to **hydrodynamic (pseudo-sound) pressure fluctuations** that stay near the surface and decay away (not propagating as acoustic waves)
- At **low frequencies**, $k_i^{(o)} \rightarrow 0$, so $\Delta\tilde{p}(k_1^{(o)}, k_3^{(o)}, \omega)$ reduces to the net unsteady loading on the airfoil, $\Delta\tilde{p}(\mathbf{y}, \omega)/(2\pi)^2$
- In contrast, at **very high frequencies** the wavenumber transform $\Delta\tilde{p}$ will be dominated by **edge effects** as the *integrand's oscillatory nature* tends to *suppress* the contributions from the surface where pressure varies smoothly

Wavenumber and Fourier transforms

Dipole

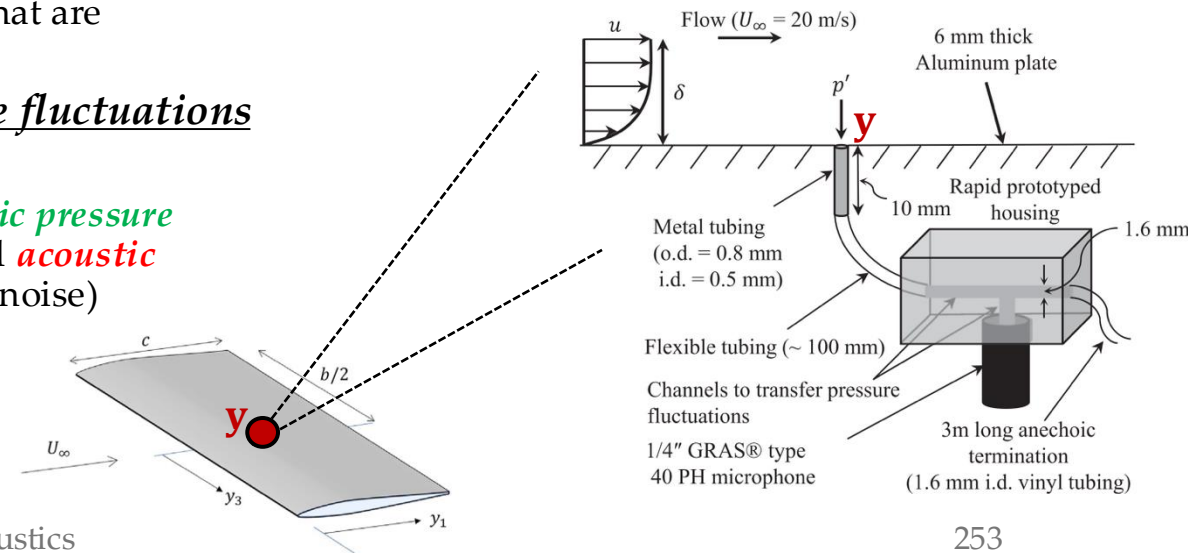
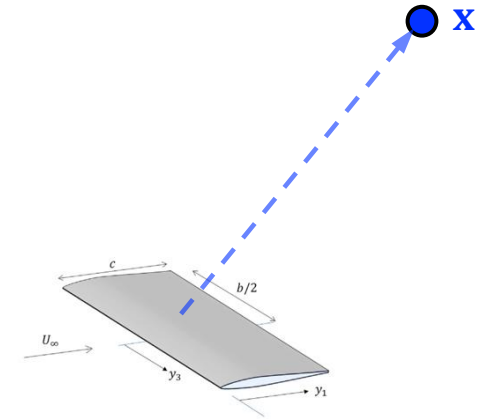
Application to thin airfoils

$$k_i^{(o)} = \frac{\omega x_i}{c_\infty |\mathbf{x}|} = k \frac{x_i}{|\mathbf{x}|}$$

Can we measure $\Delta\tilde{p}(k_1^{(o)}, k_3^{(o)}, \omega)$ on the airfoil and associate them with the acoustic far-field?

- Easiest approach – far-field sound measurements for values of $k_1^{(o)}$ & $k_3^{(o)}$ in the range $-\omega/c_\infty < k_i^{(o)} < \omega/c_\infty$ (for a range of observer angles and frequencies)
 - Thus, the 2D wavenumber spectrum of the pressure jump $\Delta\tilde{p}(k_1^{(o)}, k_3^{(o)}, \omega)$ can be obtained at all the wavenumbers that are **important for the acoustic calculations**
- Challenging approach – measuring the surface pressure fluctuations and relate them to the acoustic far-field
 - Measurement needs to **distinguish** between **hydrodynamic pressure fluctuations**, convected subsonically over the surface, and **acoustic fluctuations** (which also may be contaminated by facility noise)

$$(\tilde{p}(\mathbf{x}, \omega) c_\infty^2)_{\text{dipole}} \approx \frac{-i\pi\omega x_2 \Delta\tilde{p}(k_1^{(o)}, k_3^{(o)}, \omega) e^{ik|\mathbf{x}|}}{c_\infty |\mathbf{x}|^2}$$



Ffowcs Williams and Hawkings (FW-H) equation

- ❑ So far, we have derived the general solution to the wave equation for a medium with stationary scattering surfaces
- ❑ **What happens when the surface is moving?**

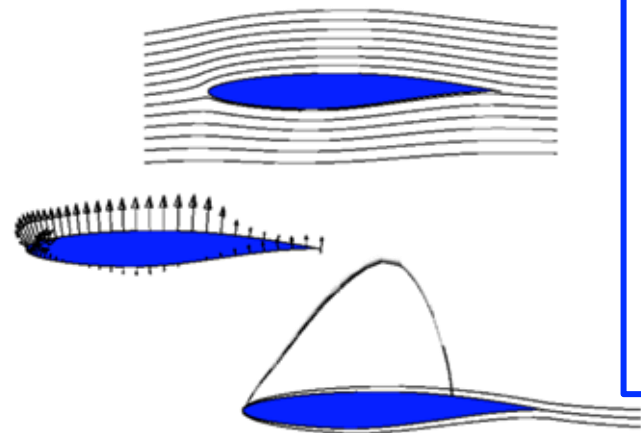


Aircraft Aeroacoustics

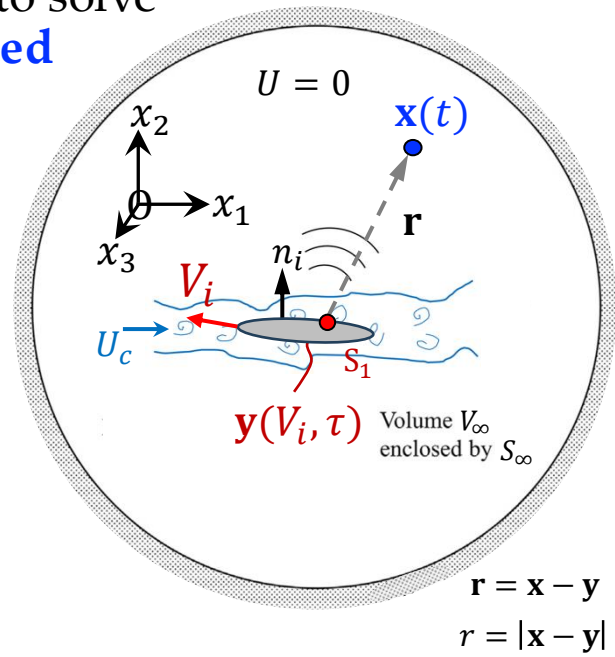


Ffowcs Williams and Hawkings (FW-H) equation

- ❑ In many aeronautical applications, the surfaces/bodies are moving (e.g., propellers and rotors, aircraft translation motion), so we need to modify our analysis to take full account of surface motion
- ❑ Powerful techniques to address this problem have been pioneered by [Ffowcs Williams and Hawkings](#)¹ (1969), as well as [Farassat](#)² (1981)
- ❑ Derivation is based on the concept of **generalized derivatives**, which is used to solve Lighthill's equation for a medium that includes **moving surfaces and convected turbulent flow**



$$\begin{aligned} \rho'(\mathbf{x}, t) c_\infty^2 H_s = & \frac{\partial}{\partial t} \int_{S_0} \left[\frac{(\rho v_j - \rho' V_j) n_j}{4\pi r |1 - M_r|} \right]_{\tau=\tau^*} dS \\ & - \frac{\partial}{\partial x_i} \int_{S_0} \left[\frac{(p_{ij} + \rho v_i (v_j - V_j)) n_j}{4\pi r |1 - M_r|} \right]_{\tau=\tau^*} dS \\ & + \frac{\partial^2}{\partial x_i \partial x_j} \int_{V_0} \left[\frac{T_{ij}}{4\pi r |1 - M_r|} \right]_{\tau=\tau^*} dV(\mathbf{z}) \end{aligned}$$



SOUND GENERATION BY TURBULENCE AND SURFACES IN ARBITRARY MOTION

By J. L. FFWCS WILLIAMS AND D. L. HAWKINGS

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(Communicated by M. J. Lightfoot, Sec.R.S., Received 26 July 1968)

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Linear Acoustic Formulas for Calculation of Rotating Blade Noise

F. Farassat

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Abstract: A method is presented for the calculation of the sound field of a rotating blade in an arbitrary motion. The method is based on the use of the generalized derivatives of the surface motion. The method is applied to the case of a rotating blade in a uniform flow. The results are compared with the results of the Ffowcs Williams and Hawkings equation. The method is shown to be valid for arbitrary motions of the surface. The method is also shown to be valid for arbitrary motions of the surface. The method is also shown to be valid for arbitrary motions of the surface.

¹ Ffowcs Williams, J. E., and Hawkings, D. L. (1969) "Sound generation by turbulence and surfaces in arbitrary motion", *Phil Trans A*, 264(1151), pp. 321–342.

² Farassat, F. (1981) "Linear acoustic formulas for calculation of rotating blade noise", *AIAA Journal*, 19(9), pp. 1122–1130.

Generalized derivative - review

- ❑ A **generalized derivative** extends the concept of an ordinary derivative to **discontinuous functions**
- ❑ Consider the Heaviside step function $H_s(x)$:

$$H_s(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0, & \text{if } x < 0 \end{cases}$$

- ❑ With its derivative being: $\frac{\partial H_s(x)}{\partial x} = \begin{cases} 0, & \text{if } x > 0 \text{ and } x < 0 \\ \infty, & \text{if } x = 0 \end{cases}$

- ❑ The integral of the derivative is given by:

$$\int_{x_1}^x \frac{\partial H_s(x')}{\partial x'} dx' = \begin{cases} 1, & \text{if } x > 0 \text{ and } x_1 < 0 \\ 0, & \text{if } x < 0 \text{ and } x_1 < 0, \text{ or } x > 0 \text{ and } x_1 > 0 \end{cases}$$

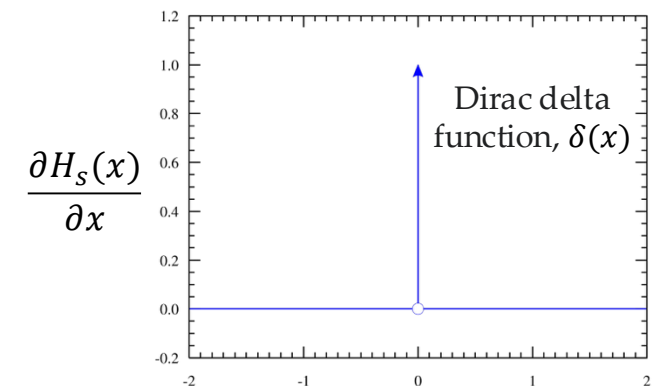
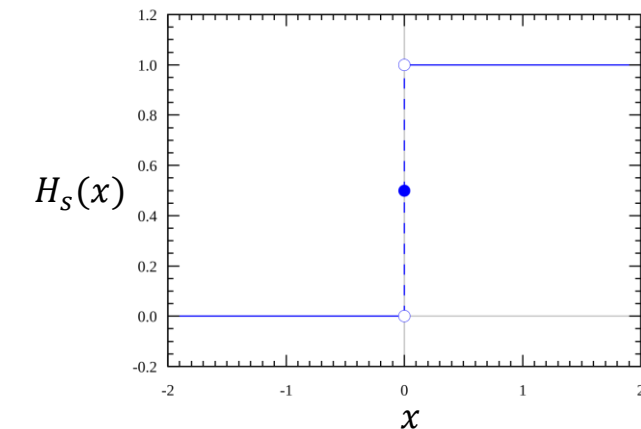
- ❑ **Looks familiar?**

➤ Derivative of $H_s(x)$ has properties of *Dirac delta function*!

$$\delta(x - x_0) = \begin{cases} \infty, & \text{if } x = x_0 \\ 0, & \text{if } x \neq x_0 \end{cases}$$

$$\int_{-\infty}^{\infty} \delta(x - x_0) dx = \begin{cases} 1, & \text{if } x = x_0 \\ 0, & \text{if } x \neq x_0 \end{cases}$$

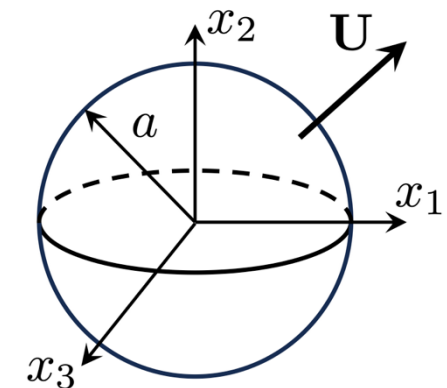
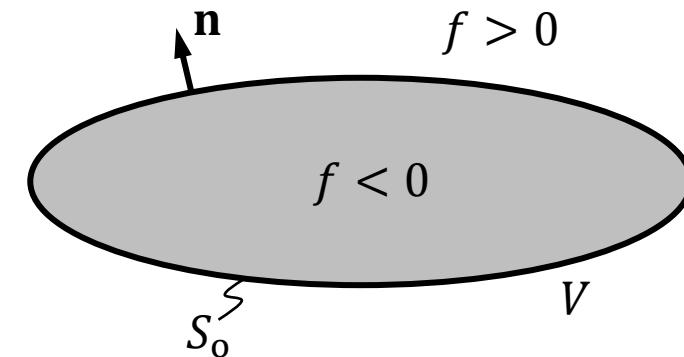
$$\delta(x - x_0) = \frac{\partial H_s(x - x_0)}{\partial x}$$



Generalized derivative - review

- ❑ We can extend this concept to surfaces that bound a region
- ❑ Consider a function $f(\mathbf{x})$ that is greater than zero outside a volume V enclosed by a surface S_0 and less than zero inside – therefore, surface defined by $f = 0$
 - Spherical surface of radius a : $f = x_1^2 + x_2^2 + x_3^2 - a^2$
 - Cylinder of radius a : $f = x_1^2 + x_2^2 - a^2$
- ❑ Unit normal vector pointing out of surface: $\mathbf{n} = \frac{\nabla f}{|\nabla f|}$ @ $f = 0$
- ❑ **Surface description is general** and can be extended to **moving surfaces** by letting f also be a function of time – $f(\mathbf{x}, t)$
 - For example, a **sphere moving with velocity** $\mathbf{U}$ is defined by:

$$f(\mathbf{x}, t) = |\mathbf{x} - \mathbf{U}t|^2 - a^2$$

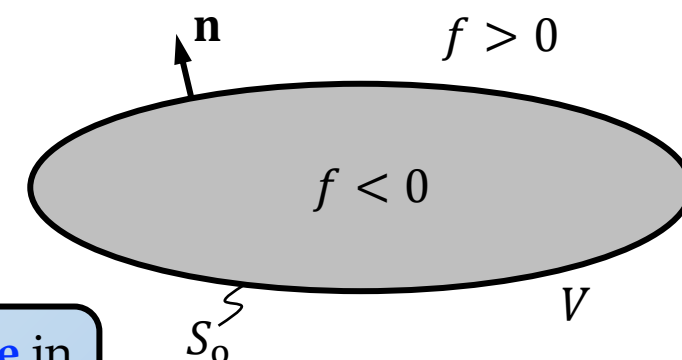


Generalized derivative – towards moving surface

- ❑ To analyze the flow field in the region exterior to a surface S_0 , which may be in **arbitrary accelerated motion**, we need information about the *flow variables* in the region exterior to S_0
 - Note that we know *nothing* about the flow variables inside S_0
- ❑ Thus, for example, **flow velocity can only be defined outside S_0**
- ❑ Introducing a *new flow velocity variable* $\mathbf{v}H_s(f)$:

$$\mathbf{v}H_s(f) = \begin{cases} \mathbf{v}, & \text{if } f > 0 \\ 0, & \text{if } f < 0 \end{cases}$$

We now have a **flow variable that is defined everywhere** in the presence of an **arbitrary moving surface**



- ❑ **Divergence** of this new function: $\nabla \cdot (\mathbf{v}H_s(f)) = H_s(f)\nabla \cdot \mathbf{v} + \mathbf{v} \cdot \nabla H_s(f)$



$$\nabla \cdot (\mathbf{v}H_s(f)) = H_s(f)\nabla \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{n}\delta(f)|\nabla f|$$

$$\nabla H_s(f) = \frac{\partial H_s}{\partial f} \nabla f = \delta(f)|\nabla f| \mathbf{n}$$

$$\mathbf{n} = \frac{\nabla f}{|\nabla f|}$$

$$\delta(x) = \frac{\partial H_s(x)}{\partial x}$$

Generalized derivative – towards moving surface

- Integrate over an infinite region V_∞ bounded at infinity by the surface S_∞ , and use *divergence theorem*:

$$\int_{V_\infty} \nabla \cdot (\mathbf{v} H_s(f)) dV = - \int_{S_\infty} (\mathbf{v} H_s(f)) \cdot \mathbf{n} dS \implies \int_{V_\infty} (H_s(f) \nabla \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{n} \delta(f) |\nabla f|) dV = - \int_{S_\infty} \mathbf{v} \cdot \mathbf{n} dS \quad (\text{I})$$

$$H_s(f) = 1 \text{ on } S_\infty$$

$$\nabla \cdot (\mathbf{v} H_s(f)) = H_s(f) \nabla \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{n} \delta(f) |\nabla f|$$

- For the **first term** in the volume integral, we recognized $H_s = 0$ inside the body, and so we can restrict the integration to the **exterior volume region** V_0 outside of S_0 ($f > 0$) and apply the divergence theorem:

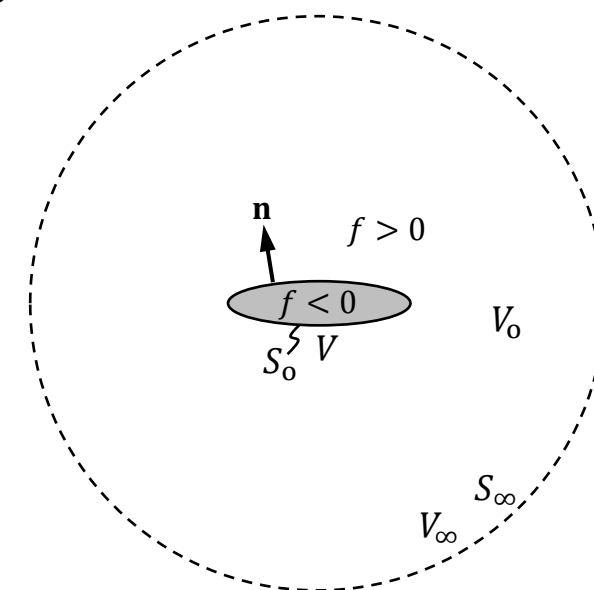
$$\int_{V_\infty} (H_s(f) \nabla \cdot \mathbf{v}) dV = \int_{V_0} (\nabla \cdot \mathbf{v}) dV = - \int_{S_0 + S_\infty} \mathbf{v} \cdot \mathbf{n} dS \quad (\text{II})$$

Combining (I) and (II)

$$\int_{V_\infty} (\mathbf{v} \cdot \mathbf{n} \delta(f) |\nabla f|) dV = - \int_{S_0} \mathbf{v} \cdot \mathbf{n} dS$$

$$\nabla H_s(f) = \delta(f) |\nabla f| \mathbf{n}$$

$$H_s(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0, & \text{if } x < 0 \end{cases}$$



Key result – Dirac delta distribution $\delta(f) |\nabla f|$ turns a volume integral over all space, where the surface motion is encoded implicitly by $f(\mathbf{x}, t) = 0$, into an **equivalent** surface integral over the moving body surface $S_0(t)$

Generalized derivative – towards moving surface

- Now, consider the time derivative of the *new flow velocity variable* $\mathbf{v}H_s(f)$:

$$\frac{\partial(\mathbf{v}H_s(f))}{\partial t} = H_s(f) \frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \delta(f) \frac{\partial f}{\partial t}$$

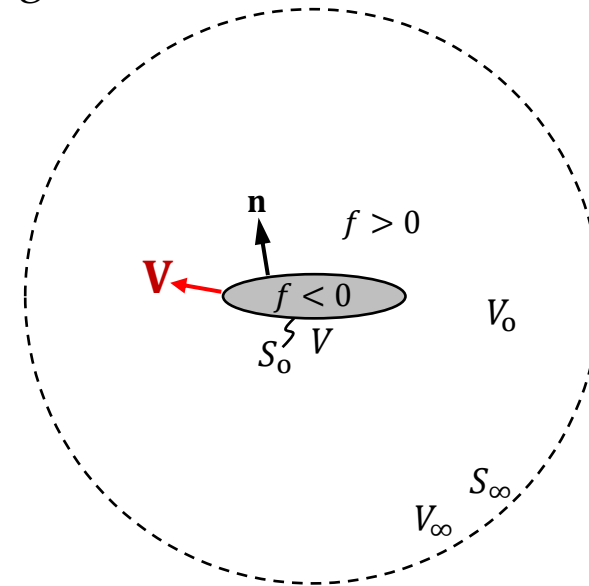
$$\delta(t) = \frac{\partial H_s(t)}{\partial t}$$

- For a surface defined by $f(\mathbf{x}, t)$ moving with velocity $\mathbf{V}$, it follows that $f(\mathbf{x} - \int \mathbf{V} dt)$ on the surface, and so the chain rule on the surface ($Df/Dt = \partial f/\partial t + \mathbf{V} \cdot \nabla f = 0$, since $f = 0$) gives:

$$\left[\frac{\partial f(\mathbf{x} - \int \mathbf{V} dt)}{\partial t} \right]_{f=0} = [-\mathbf{V} \cdot \nabla f]_{f=0}$$

$$\Rightarrow \frac{\partial(\mathbf{v}H_s(f))}{\partial t} = H_s(f) \frac{\partial \mathbf{v}}{\partial t} - \mathbf{v} \delta(f) (\mathbf{V} \cdot \mathbf{n} |\nabla f|) \quad \mathbf{n} = \frac{\nabla f}{|\nabla f|}$$

- So the second term on the right is specified in terms of the surface velocity $\mathbf{V}$



Generalized derivative – towards moving surface

- Similarly, for a scalar variable, such as the density perturbation ρ' , we find:

$$\frac{\partial(\rho' H_s(f))}{\partial t} = H_s(f) \frac{\partial \rho'}{\partial t} - \rho' \delta(f) (\mathbf{V} \cdot \mathbf{n} |\nabla f|)$$

$$\delta(x) = \frac{\partial H_s(x)}{\partial x}$$

$$\frac{DH_s(f)}{Dt} = \frac{\partial H_s}{\partial f} \frac{Df}{Dt} = \delta(f) \frac{Df}{Dt}$$

- Consider the material derivative of $\mathbf{v} H_s(f)$ given by:

$$\frac{D(\mathbf{v} H_s)}{Dt} = \frac{D\mathbf{v}}{Dt} H_s + \underbrace{\mathbf{v} \frac{Df}{Dt}}_{\text{Should be 0 for a solid surface}} \delta(f)$$

Should be 0 for a solid surface

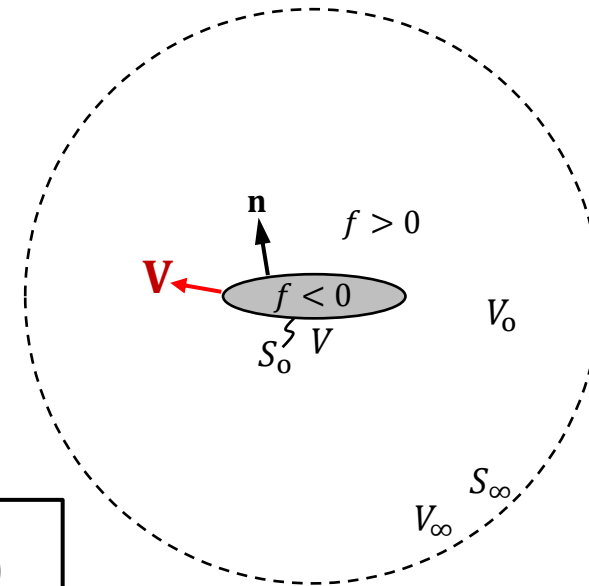
➤ **BUT!** $\frac{Df}{Dt} = \frac{\partial f}{\partial t} + (\mathbf{v} \cdot \nabla) f \implies \frac{Df}{Dt} = (\mathbf{v} - \mathbf{V}) \cdot \mathbf{n} |\nabla f|$

$\frac{\partial f}{\partial t} = -\mathbf{V} \cdot \mathbf{n} |\nabla f|$

- For an **impermeable (solid) surface** – $\mathbf{v} \cdot \mathbf{n} = \mathbf{V} \cdot \mathbf{n}$

➤ Fluid velocity normal to the surface is equal to the surface velocity

$$\frac{Df}{Dt} = 0$$



Generalized derivative – towards moving surface

□ Looking ahead, we will deal with terms of the form:

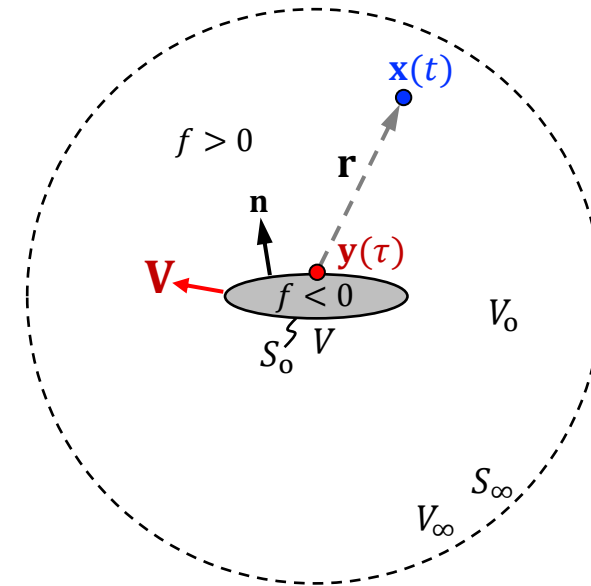
$$\begin{aligned} \int_{V_\infty} G(\mathbf{x}, t | \mathbf{y}, \tau) \frac{\partial}{\partial y_i} (p_{ij} n_j \delta(f) |\nabla f|) dV(\mathbf{y}) &= \underbrace{\int_{V_\infty} \frac{\partial}{\partial y_i} (G(\mathbf{x}, t | \mathbf{y}, \tau) p_{ij} n_j \delta(f) |\nabla f|) dV(\mathbf{y})}_{\text{Sommerfeld radiation condition}} - \int_{V_\infty} (p_{ij} n_j \delta(f) |\nabla f|) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial y_i} dV(\mathbf{y}) \\ &= - \int_{S_\infty} (G(\mathbf{x}, t | \mathbf{y}, \tau) p_{ij} n_j \delta(f) |\nabla f|) \cdot \mathbf{n} dS = 0 \end{aligned}$$

Sommerfeld radiation condition

$$\Rightarrow \int_{V_\infty} G(\mathbf{x}, t | \mathbf{y}, \tau) \frac{\partial}{\partial y_i} (p_{ij} n_j \delta(f) |\nabla f|) dV(\mathbf{y}) = - \int_{V_\infty} (p_{ij} n_j \delta(f) |\nabla f|) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial y_i} dV(\mathbf{y})$$

□ Similarly, for integrands that include time derivatives, we can show that integrating by parts and noting that G is zero when $\tau = \pm T$, then:

$$\int_{-T}^T \int_{V_\infty} G(\mathbf{x}, t | \mathbf{y}, \tau) \frac{\partial}{\partial \tau} (v_j n_j \delta(f) |\nabla f|) dV(\mathbf{y}) d\tau = - \int_{-T}^T \int_{V_\infty} (v_j n_j \delta(f) |\nabla f|) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial y_i} dV(\mathbf{y})$$



The FW-H equation

- We now show how the concept of **generalized derivatives** may be used to obtain the solution to Lighthill's equation in the presence of moving surfaces
- We start by evaluating the **continuity and momentum equations** in terms of the new variables $\mathbf{v}H_s$, pH_s , and ρH_s , which are defined everywhere in an unbounded infinite volume V_∞
- To obtain the continuity equation in terms of the new variables, we expand:

$$\frac{\partial(\rho'H_s)}{\partial t} + \frac{\partial(\rho v_i H_s)}{\partial x_i} = \underbrace{H_s \left(\frac{\partial \rho'}{\partial t} + \frac{\partial(\rho v_i)}{\partial x_i} \right)}_{\text{So, 0 everywhere}} + (\rho v_j - \rho' V_j) n_j \delta(f) |\nabla f|$$

Outside flow
 $\frac{\partial \rho'}{\partial t} + \frac{\partial(\rho v_i)}{\partial x_i} = 0$ Continuity eq. (in flow region)

Continuity eq. →

$$\frac{\partial(\rho'H_s)}{\partial t} + \frac{\partial(\rho v_i H_s)}{\partial x_i} = (\rho v_j - \rho' V_j) n_j \delta(f) |\nabla f|$$

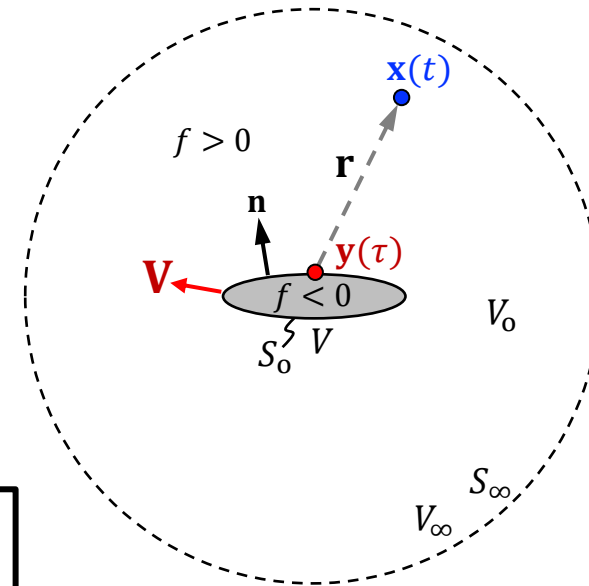
similarly...

Momentum eq. →

$$\frac{\partial(\rho v_i H_s)}{\partial t} + \frac{\partial(\rho v_i v_j H_s + p_{ij} H_s)}{\partial x_j} = (\rho v_i (v_j - V_j) + p_{ij}) n_j \delta(f) |\nabla f|$$

$$p_{ij} = (p - p_\infty) \delta_{ij} - \sigma_{ij}$$

$$\begin{aligned} \frac{\partial H_s(f)}{\partial x_i} &= \frac{\partial H_s}{\partial f} \nabla f = \delta(f) |\nabla f| \mathbf{n} \\ \frac{\partial H_s(f)}{\partial t} &= \frac{\partial H_s}{\partial f} \frac{\partial f}{\partial t} = \delta(f) \frac{\partial f}{\partial t} \\ \frac{\partial f}{\partial t} &= -\mathbf{V} \cdot \mathbf{n} |\nabla f| \\ \rho' &= \rho - \rho_\infty \end{aligned}$$



The FW-H equation

□ To obtain a wave equation, follow the same procedure as for Lighthill's equation...

□ Continuity eq.
$$\frac{\partial}{\partial t} \left(\frac{\partial(\rho' H_s)}{\partial t} + \frac{\partial(\rho v_i H_s)}{\partial x_i} \right) = \frac{\partial}{\partial t} \left((\rho v_j - \rho' V_j) n_j \delta(f) |\nabla f| \right)$$

□ Momentum eq.
$$\frac{\partial}{\partial x_i} \left(\frac{\partial(\rho v_i H_s)}{\partial t} + \frac{\partial(\rho v_i v_j H_s + p_{ij} H_s)}{\partial x_j} \right) = \frac{\partial}{\partial x_i} \left((\rho v_i (v_j - V_j) + p_{ij}) n_j \delta(f) |\nabla f| \right)$$

(–)
Subtract

Take the result and subtract $\partial^2(\rho' c_\infty^2 H_s) / \partial x_i^2$

➔
$$\frac{\partial^2(\rho' H_s)}{\partial t^2} - c_\infty^2 \frac{\partial^2(\rho' H_s)}{\partial x_i^2} = \frac{\partial^2(T_{ij} H_s)}{\partial x_i \partial x_j} - \frac{\partial}{\partial x_i} \{ (\rho v_i (v_j - V_j) + p_{ij}) n_j \delta(f) |\nabla f| \} + \frac{\partial}{\partial t} \{ (\rho v_j - \rho' V_j) n_j \delta(f) |\nabla f| \}$$

FW-H equation

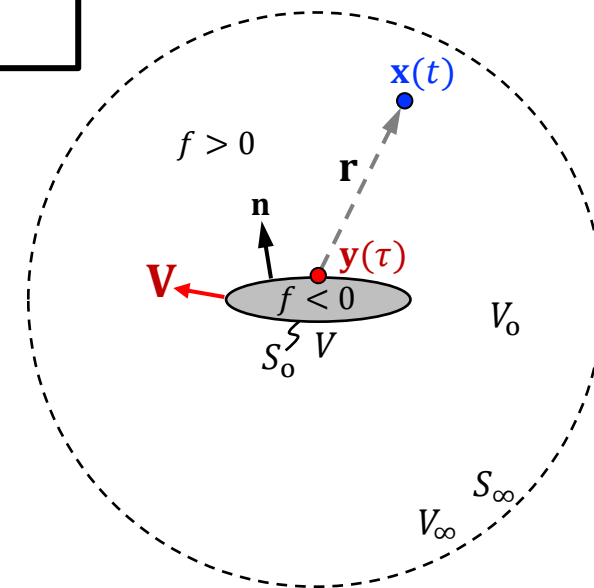
Lighthill's stress tensor $T_{ij} = \rho v_i v_j + [(p - p_\infty) - (\rho - \rho_\infty) c_\infty^2] \delta_{ij} - \sigma_{ij}$

Solution to FW-H analogy

- A solution to the FW-H analogy can be obtained by using the method of Green's functions (as before)
- We start by writing FW-H analogy in terms of $\mathbf{y}$ and τ as follows:

$$\frac{\partial^2(\rho' H_s)}{\partial \tau^2} - c_\infty^2 \frac{\partial^2(\rho' H_s)}{\partial \mathbf{y}_i^2} = \mathcal{S} = \frac{\partial^2(T_{ij} H_s)}{\partial \mathbf{y}_i \partial \mathbf{y}_j} - \frac{\partial}{\partial \mathbf{y}_i} \{(\rho v_i (v_j - V_j) + p_{ij}) n_j \delta(f) |\nabla f|\} \\ + \frac{\partial}{\partial \tau} \{(\rho v_j - \rho' V_j) n_j \delta(f) |\nabla f|\}$$

$$T_{ij} = \rho v_i v_j + [(p - p_\infty) - (\rho - \rho_\infty) c_\infty^2] \delta_{ij} - \sigma_{ij}$$



Solution to FW-H analogy

- Since the dependent variable ($\rho' H_s$) is defined in an unbounded medium, we don't need to deal with the surface integrals and can just use a **simple convolution**:

$$\begin{aligned} \rho'(\mathbf{x}, t) c_\infty^2 H_s = & \int_{-T}^T \int_{V_\infty} G \left(\frac{\partial^2 (T_{ij} H_s)}{\partial y_i \partial y_j} \right) dV(\mathbf{y}) d\tau - \int_{-T}^T \int_{V_\infty} \left(G \frac{\partial}{\partial y_i} \{ (\rho v_i (v_j - V_j) + p_{ij}) n_j \delta(f) |\nabla f| \} \right) dV(\mathbf{y}) d\tau \\ & + \int_{-T}^T \int_{V_\infty} \left(G \frac{\partial}{\partial \tau} \{ (\rho v_j - \rho' V_j) n_j \delta(f) |\nabla f| \} \right) dV(\mathbf{y}) d\tau \end{aligned}$$

- Providing $\mathbf{x}$ is inside V and $-T < t < T$